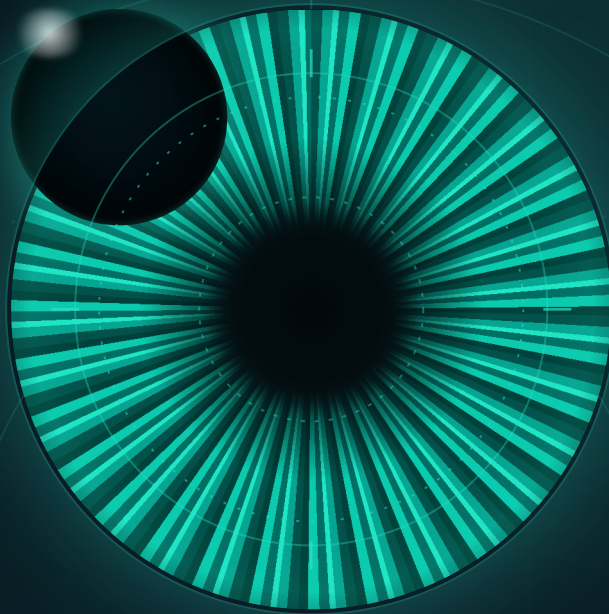




# EyeBot

*An AI training companion that turns SNEC's OAOT curriculum into a living tutor, an unlimited clinical-simulation suite, and a self-optimising memory engine.*



PREPARED FOR THE 2026 INTERNATIONAL E-LEARNING AWARDS · ACADEMIC DIVISION ·  
E-LEARNING EXPERIENCES

Every ophthalmic allied-health professional deserves a tutor who never sleeps, a patient who is always available, and a curriculum that adapts to exactly what they keep forgetting.

**SNEC EyeBot** is an AI-powered learning companion built on SNEC's own **OAOT Module 1** curriculum. It unites three evidence-based modalities — a Socratic AI tutor, interactive clinical case simulations, and spaced-repetition flashcards — into one continuous loop that personalises itself to each learner. Running quietly alongside is an analytics layer that gives educators, for the first time, a live view of where an entire cohort is thriving or struggling. The platform is fully engineered, demonstrable today, and ready for a SNEC pilot.

## 3-in-1



Tutor, clinical simulator and memory engine working as a single adaptive loop.

## 100%



Curriculum-grounded — answers drawn from SNEC's OAOT Module 1, not the open internet.

## Cohort



Live at-risk detection, retention scoring and weekly digests for educators.

A fixed curriculum becomes a **living, personalised, measurable** learning experience — and positions SNEC at the front of AI-enabled clinical education.

# Training the eyes behind the eye care.

Ophthalmic Assistants, Technicians and Patient Service Associates are the first clinical touchpoint at SNEC. Their training is rigorous — and rightly so — but it meets four structural limits that no amount of dedication can fully solve.



## Expert time is scarce

One educator cannot give every trainee unlimited one-to-one tutoring or on-demand case practice.



## Case exposure is uneven

Which emergencies a trainee actually meets is left to the luck of the clinical roster.



## Knowledge decays

Without deliberate reinforcement, much of what is taught is forgotten within weeks of the lesson.

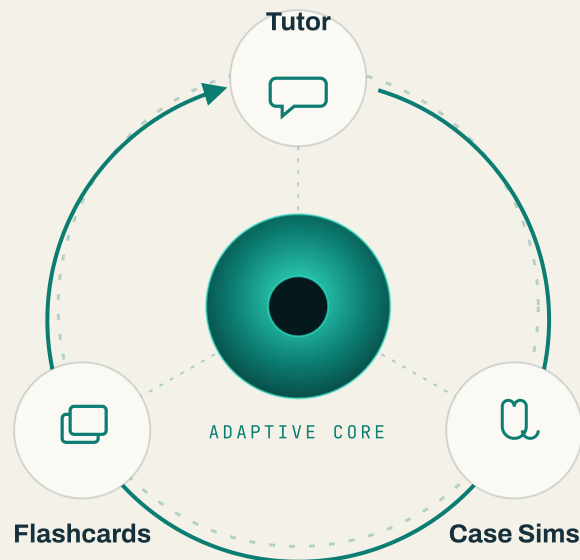


## Cohort gaps are invisible

Educators often discover weaknesses at assessment time — when it is already late to intervene.

EyeBot is engineered to dissolve all four — **at once.**

# One companion. Three modes. A single adaptive loop.



## MODE 01

### Socratic Tutor

Plain-language questions met with Explanation → Mechanism → Clinical Pearl → a check for understanding, always grounded in the OAOT knowledge base.



## MODE 02

### Case Simulations

Interview a virtual patient, order investigations, commit to a diagnosis and plan — graded on a four-domain rubric out of 40 with instant feedback.



## MODE 03

### Spaced Flashcards

High-yield cards auto-generated from each session and every missed clinical point, scheduled by the SM-2 algorithm for durable recall.

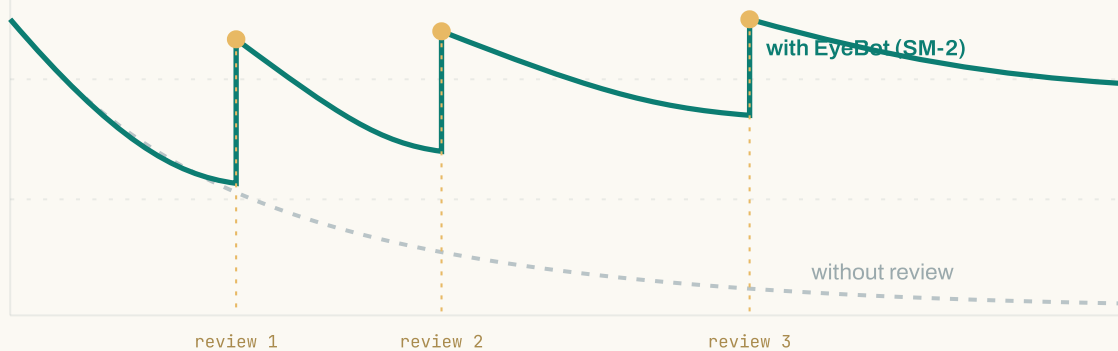
...and quietly, an analytics layer watches the whole cohort — turning individual practice into institutional insight.

# Grounded in learning science, not novelty.

Every design decision in EyeBot maps to an established principle of how clinicians actually learn and retain.

FIG. 1 — SPACED REPETITION VS. THE FORGETTING CURVE

Retention ↑ · Time →



## Retrieval & spacing

SM-2 spaced repetition fights the forgetting curve — each review lengthens durable retention.



## Case-based reasoning

Simulations rehearse the full clinical chain: history, investigation, diagnosis, management.



## Socratic scaffolding

The tutor explains mechanism, then checks understanding — active recall over passive reading.



## Formative feedback

Rubric scores and targeted cards turn every mistake into the learner's next rep.

# Far more than a chatbot.



## Curriculum-grounded

Retrieval over SNEC's OAOT Module 1 — clinically faithful answers, not generic AI guesswork.



## Self-optimising

Weak topics automatically become tomorrow's flashcards and tougher simulated cases.



## Role-aware

Tailored content and case libraries for the OA, OT and PSA training pathways.



## Educator analytics

Cohort heatmap, at-risk detection, learning-velocity & retention scoring, weekly e-mail digests.



## Governance built-in

PDPA-compliant consent, bcrypt-secured credentials and role-based access — privacy by design.



## Reliable by architecture

A Workflows–Agents–Tools design separates AI reasoning from deterministic execution.

### WORKFLOWS

plain-language clinical SOPs

### AGENTS

AI orchestration & judgement

### TOOLS

deterministic, tested code

# From prototype to proof.

## PROPOSED SNEC PILOT

### COHORT

One OAOT intake over an 8–12 week module, benchmarked against a prior cohort as historical control.

### METRICS

Pre/post knowledge test · case-rubric progression · flashcard retention · engagement (active days & streaks) · educator time saved · learner satisfaction.

### METHOD

Baseline → guided daily use → measured uplift, with the analytics layer capturing evidence automatically throughout.

## FIT TO IELA CRITERIA

### ATTRIBUTE

### HOW EYEBOT DELIVERS

#### Educational soundness

##### PEDAGOGY

Curriculum-grounded; SM-2 spacing; Socratic + case-based design rooted in learning science.

#### Effectiveness

##### OUTCOMES

Closes gaps in a measured loop; analytics quantify uplift at learner and cohort level.

#### User experience

##### ENGAGEMENT

Polished, gamified and mobile-friendly, with daily check-ins that build habit.

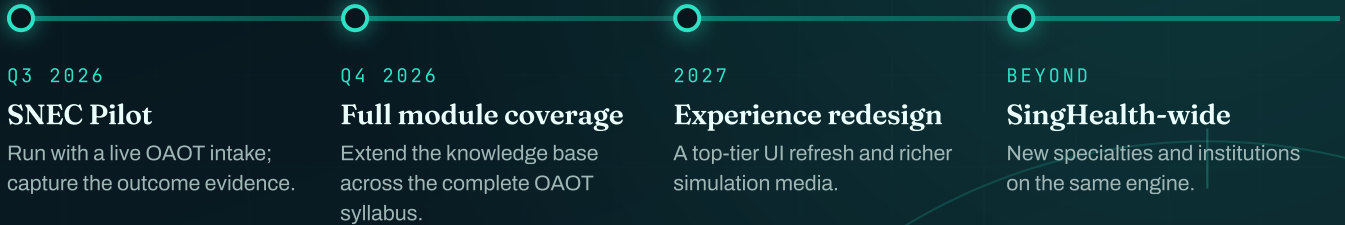
#### Significance

##### IMPACT

First AI companion for ophthalmic allied-health training; scalable across SingHealth modules.

Because impact is **measured, not asserted**, the pilot doubles as the evidence base for the award submission itself.

# Where this goes next.



SNEC EyeBot turns a fixed curriculum into a **living, personalised and measurable** learning experience — and positions SNEC at the forefront of AI-enabled clinical education.

ENGINEERED WITH

FastAPI

Supabase

React + TypeScript

Anthropic Claude

Workflows · Agents · Tools